



Lesson Aim

Understand the importance of correct storage and how it prevents spoilage and enhances maturity.

INTRODUCTION

Some wines are best left to mature, and drunk when many years old; but others are better drunk when young.

Wines that require longer to be matured are more costly, the cost of a longer storage being offset by the increased price.

Most of the lighter white wines, which are popular in modern times will taste best when relatively young. Over time their fresh, fruity flavours tend to fade.

Once a wine has gone matured, stabilised and clarified it is commonly stored in either, inert containers (e.g. stainless steel, fibreglass or concrete) or wood casks (e.g. oak). It might also be stored in bottles.

- Inert containers do not alter the taste to any significant extent provided contact with air is prevented.
- Oak casks can enhance/change the taste.
- Wines are often kept in large inert containers or casks for a time, then bottled.

INERT CONTAINER STORAGE

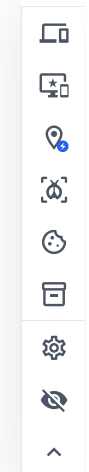
Wine can be stored like this for months with minimal and non-significant changes provided it is:

- Stored at 10-15°C.
- A layer of carbon dioxide or nitrogen is kept over the wine surface to prevent exposure to air.
- Sulphur dioxide levels are held at 30-40 ppm.

STORAGE/ AGING IN OAK CASKS OR BARRELS

This is more complicated:

- Wood can affect flavour, but not all wines are improved by adding an oak or wood flavour.
- Flavours embedded in the wood from previous use, will affect the current vintage.
- Different species of oak will create different flavours in the stored wine.
- The length of time stored in wood is critical (It needs to be long enough to impart the flavour, but not so long that the oak flavour becomes overpowering).
- There can be a small, slow exchange of gasses from the cellar atmosphere through the wood, into and out of the cask.



- The cask is commonly sealed with a wooden plug (called a bung). If the plug is exposed to air, evaporation from the wine inside the cask will increase, and an air pocket is more prone to developing below the plug. This issue is minimised by rotating the barrel after plugging the top, so the plug sits to the side and remains totally covered by liquid.

Red wines are generally more suited to storing and aging in wood casks, though some distinct types of white wines may also be suitable to keep in a cask.

Managing wine aging and storage, particularly in wood casks, is a complex task that requires practice, experimentation, and experience to perfect. The winemaker needs to monitor the environment and the wine carefully to keep the environmental conditions right and take appropriate decisions about when to tap and bottle the wine from the cask.

MATURING AND STORING IN BOTTLES

Wine from oak casks or inert containers, which is judged to be mature, is transferred to bottles. Some wines may be sold as soon as they are bottled, but others are kept for a further period of maturing before marketing.

Aging wine in a bottle will change it. Some characteristics will fade and eventually be lost; but as that happens, other characteristics will develop and strengthen. When wines are aged in a bottle its components, including alcohol, acidity, flavour, and sharpness, do change over time.

Bottled wine is best stored in low humidity, cool, dark conditions and away from vibration (e.g. machinery or vehicles nearby). Corked bottles are laid on the side, so liquid always covers the cork, to minimise air getting into the bottle.

Time allocated to bottle aging depends upon:

- Techniques/processes the wine went through before bottling
- Wine style required
- Taste preference of target market
- Temperature – wine stored at lower temperatures will age more slowly. Generally, 10 to 15°C is considered best for most wines.

Wine that is aged in bottles should be clear and balanced in its main components.

Quality white and rose' wines are generally given 3 to 5 years to mature. Certain white wines are considered better after even longer periods in the bottle (including Australian Chardonnay). Sparkling wines are ready to drink when bottled, and quality may diminish if stored too long.

The length of time recommended for storing red wines varies a lot. Lighter reds (e.g. Cabernet and Shiraz) may only need three to five years, but others much longer.

SPOILAGE

When a fermented drink is sealed off to the air, it is separated not only from microorganisms that might cause it to rapidly change (spoil); but the supply of oxygen that might feed "spoiling" microorganisms will be excluded.

Sulphur dioxide (SO²), a preservative, is also sometimes added to wine or other brewed beverages. This is a chemical that wine yeasts are relatively tolerant of, but undesirable microorganisms do not like at all.

To reduce any chance of spoilage winemakers will commonly follow three protocols:

- Strict hygiene
- Minimising oxygen coming in contact with the wine
- Using a preservative (sulphur dioxide).

In today's health-conscious society there is a demand for preservative-free wine, and so the use of sulphur dioxide is sometimes avoided. To do so, however, can add risk and reduce the life of the wine.

Causes of Spoilage

Spoilage needs to be guarded against. Wine quality can deteriorate or spoil due to many things, including:

- Oxidation
- Woodiness



- Mercaptans and Hydrogen Sulphide
- Yeast
- Bacteria and other microorganisms
- Turning to Vinegar

Oxidation

Oxidation can occur particularly with wines stored a long time in oak casks. An early indicator is the wine colour changing colour (browning). This then progresses to development of an increasingly bitter taste. Oxidation can be countered by increasing sulphur dioxide level and/or topping up to reduce any air trapped in a container.

Minimally spoiled wine may be salvaged using blending, or treatment with charcoal or PVPP to reduce colour.

Woodiness

Woodiness can result from excessive exposure to wood. Regular tasting can be important to avoiding this.

Hydrogen Sulphide and Mercaptans

Hydrogen Sulphide and mercaptans are issues are readily detected by an increasing sulphur odour or scent like decaying organic material. Some yeast varieties are more likely to be associated with this problem than others. This issue may be caused by high residues of sulphur sprayed on the grapes. It can also be caused by low nutrient levels in grapes or a higher pH. Wine makers should be guarded against these issues; and importantly monitor and correct pH levels if necessary.

Yeast

If wines retain a residue of 5gm or more per litre of sugar, they can be subject to yeast spoilage following fermentation. Yeast spoilage can be detected by a reduction in clarity and a release of carbon dioxide. Treat by removing wine from a barrel, fermenting dry, reducing sugar to below 2gm per litre, then restabilising, clearing, and returning to the barrel.

Bacteria

Bacterial spoilage is often detected by a white film growing on the wine surface. A slight acetic acid scent is often an early sign of bacterial spoilage. Stronger musty aromas can develop if left untreated.

Bacterial infection is more likely if pH, sulphur dioxide and/or oxygen levels drift from the ideal range.

Blending is not a good solution for bacterial infection. A better treatment is aeration, then pumping wine into inert containers (e.g. stainless steel vats) with sulphiting and a layer of injected carbon dioxide. Filtration through an activated charcoal filter can also help.

Vinegar

Acetic acid bacteria commonly known as vinegar bacteria are often responsible for the spoilage of wines through the production of acetaldehyde and acetic acid from ethanol (and glucose). They are generally described as being obligate aerobes and even short periods of exposure of the wine to oxygen, during operations such as racking and transferring, are enough to encourage significant growth of these bacteria. As oxygen gets inside the bottle and interacts with the alcohol, any wine will turn to vinegar. This happens when a cork is defective, of poor quality, or when wine is stored upright instead of on its side.

